

Week 14 Logic Lab

REASONING, DISCOVERY, AND JUDGMENT

Purpose: Apply Week 14 skills to new cases: decide what must be discovered, what can be inferred, and what judgment is needed.

Part 1: Discovery or Proof?

Question 1.1

A student says, “I proved the gym roof leaks because I made a perfect argument that old roofs often leak.” Decide what still must be discovered and what reasoning can do after that discovery.

ANSWER LABEL: Part 1 Q1

Question 1.2

A team notices that every missed deadline this month involved unclear instructions. They conclude, “Unclear instructions may be part of the problem.” Identify the discovery, the inference, and whether the conclusion is sized carefully.

ANSWER LABEL: Part 1 Q2

Part 2: Choosing the Right Intellectual Tool

Question 2.1

A debate starts over whether a school rule is “fair.” One side means fair because everyone receives the same rule; the other means fair because different needs receive different treatment. Which Week 14 tool-work is needed first, and why?

ANSWER LABEL: Part 2 Q1

Question 2.2

A club treasurer says the fundraiser succeeded because it earned more money than last year. Another student asks whether expenses also rose. What must be discovered before the conclusion is judged, and what proof question remains afterward?

ANSWER LABEL: Part 2 Q2

Part 3: Observation, Testimony, and Imagination

Question 3.1

A science group imagines that classroom plants grow faster near the window because of light. Explain why this is a useful hypothesis but not yet proof. Name the observation or method needed next.

ANSWER LABEL: Part 3 Q1

Question 3.2

A student remembers that the library was quiet during exams last year and predicts it will be quiet this year. What role does memory play here, and what testimony or observation would strengthen the judgment?

ANSWER LABEL: Part 3 Q2

Part 4: Intellectual-Tool Map in a Paragraph

Question 4.1

Read the paragraph. Make an intellectual-tool map: question, what must be discovered, what terms or claims need clarification, what inference needs testing, and what judgment is needed.

A city wants to know whether a downtown bike lane made streets safer. A cycling group says crashes “clearly dropped,” but it counts only crashes reported to the group. A driver association says traffic is worse because two morning trips took longer. The city has police crash reports, hospital injury records, and traffic-speed data, but no one has compared them yet. A council member says the lane should stay because “safer streets matter most.”

ANSWER LABEL: Part 4 Q1

Part 5: Review

Question 5.1

“The app is trustworthy because a famous athlete says it is the best study tool.” Use a prior-week source test: what authority is being used, and what would be more relevant evidence?

ANSWER LABEL: Part 5 Q1

Question 5.2

“Four students improved after using flashcards, so flashcards guarantee higher grades for every student.” Name the examples, the overclaim, and a revised claim.

ANSWER LABEL: Part 5 Q2